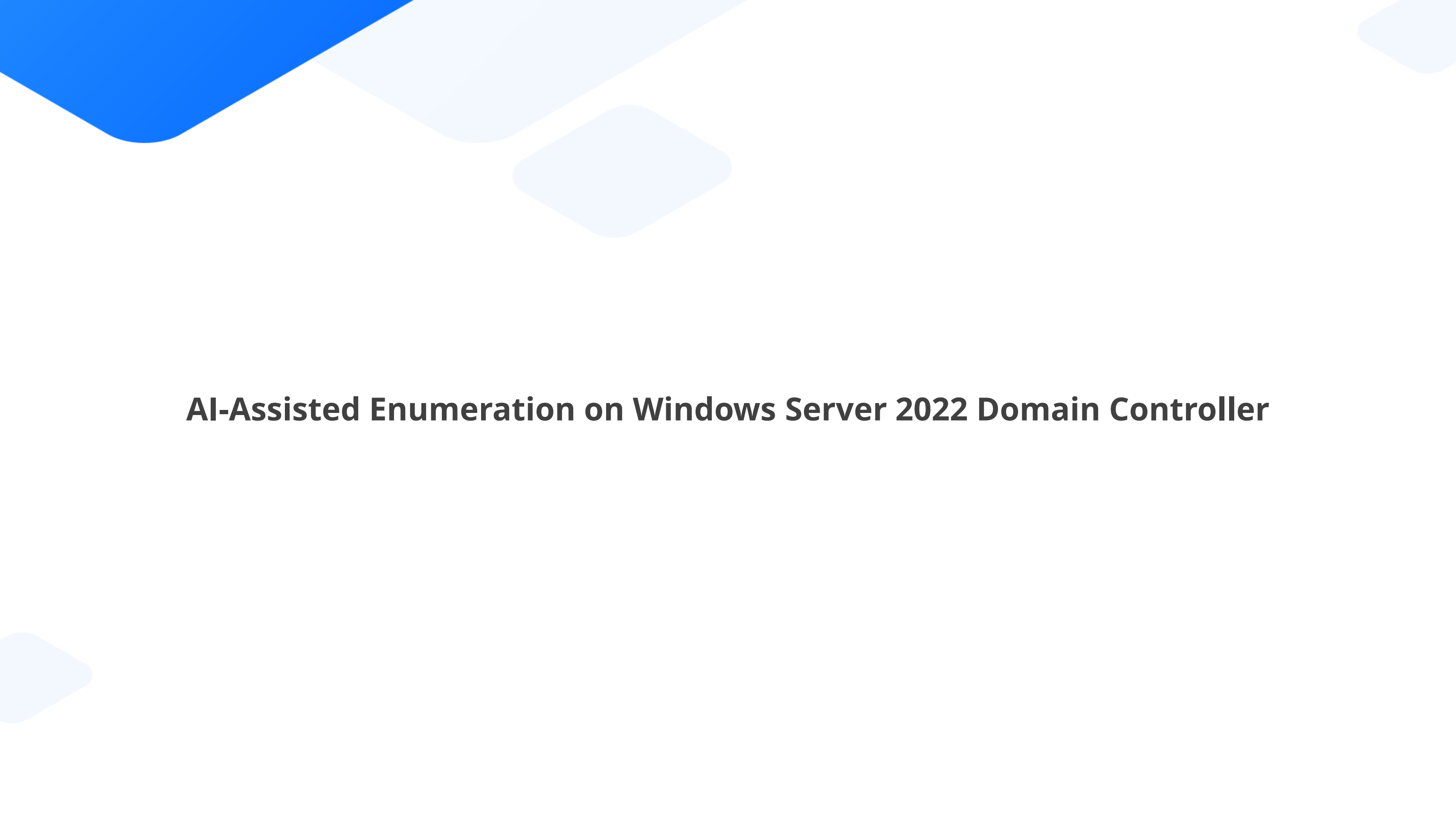


Cyber Meets AI



Capstone Project





AI-Assisted Enumeration on Windows Server 2022 Domain Controller

Objective

To simulate and analyze a multi-stage cyberattack in a controlled environment, focusing on adversary tactics, insider persistence, and investigator response

Description:

The project replicates a real-world breach scenario in which an external actor exploits poor configurations, escalates privileges, and plants persistence mechanisms. This project helps in retracing the steps of both attackers and defenders to uncover the breach lifecycle, analyze system changes, and propose mitigations. Generative AI will be integrated to support troubleshooting, simulate attacker logic, and generate forensic reports.



Problem Statement and Motivation

Real-time scenario:

TechSecure, a mid-sized enterprise, deploys Windows Server 2022 as mydomain.local, promotes it to a Domain Controller, and creates domain users while configuring SMB and AD services. From a Kali attacker VM on the same lab network, students run staged enumeration (DNS/SMB/Kerberos/LDAP) using Nmap, enum4linux, Metasploit, ADEplorer and PsTools, saving canonical outputs. ChatGPT is used throughout to craft stepwise prompts, interpret scan results, troubleshoot issues, and generate concise remediation notes and reproducible lab artifacts for assessment.



Industry Relevance

The following tools used in this project serve specific purposes within the industry:

1. **Windows Server 2022:** Enterprise-grade DC and identity platform—foundation for AD, Group Policy and centralized audit logging.
2. **Kali Linux:** Industry-standard penetration-testing OS for scanning, enumeration and validating defenses.
3. **ChatGPT:** Speeds analysis and learning—generates stepwise prompts, interprets logs, and drafts remediation.
4. **LDAP tools (ldapsearch, ADExplorer):** Inspect directory objects/ACLs to audit AD misconfigurations and map attack paths.
5. **PsTools (PsExec, PsList, etc.):** Remote process/service management for administration, forensic checks, and adversary-emulation.



Tasks

The following tasks outlines the process:

1. Query with ChatGPT
2. Rename the server
3. Configure static IP address
4. Install the AD DS
5. Promote to a domain controller
6. Troubleshoot the network problem
7. Create a user inside the DC
8. Configure the SMB with the help of AI
9. Enumerate DNS/SMB with AI
10. Enumerate SMB and Kerberos with AI
11. Enumerate LDAP using tools



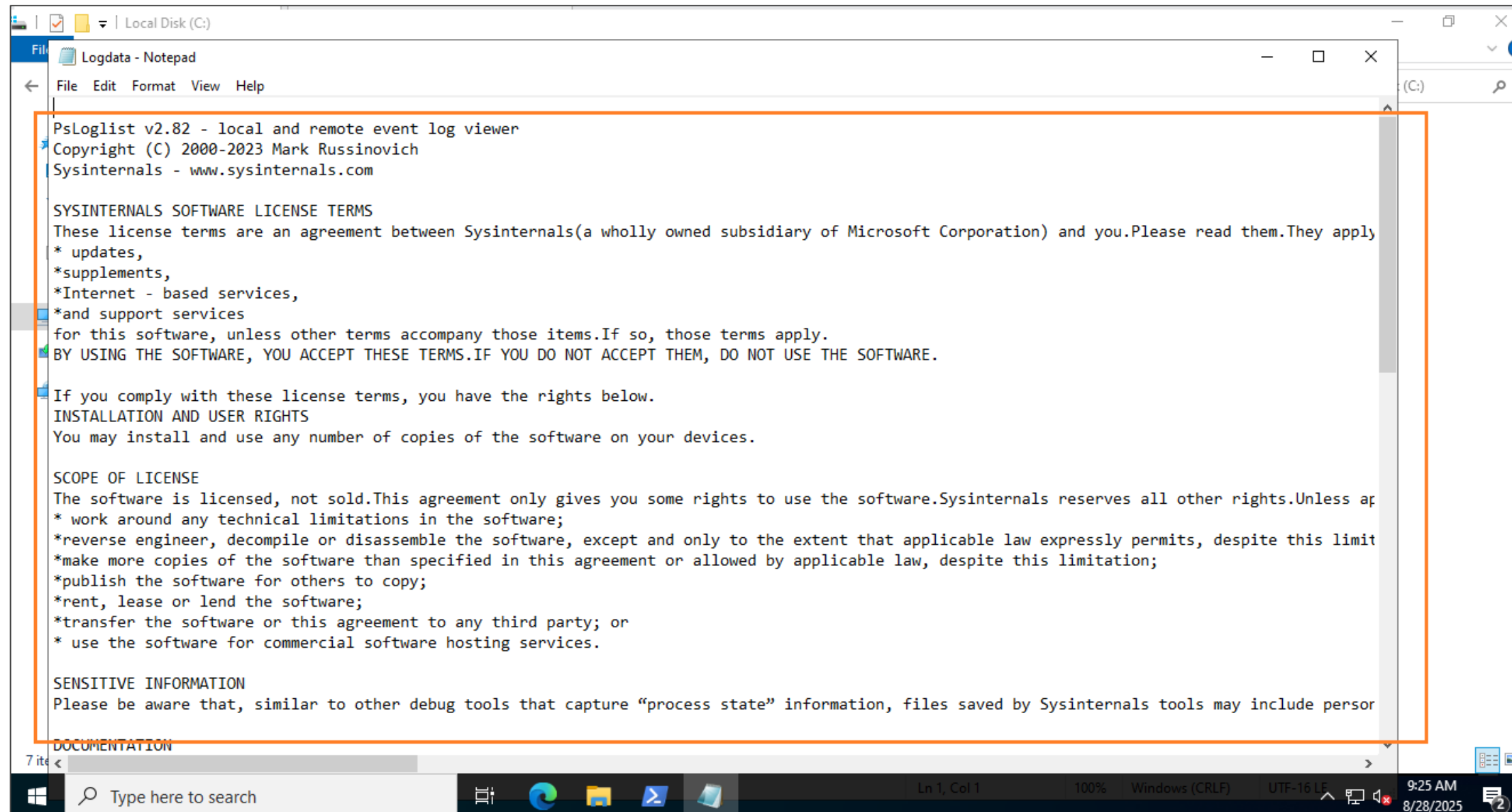
Project References



- Task 1: AI Foundations for Cybersecurity Professionals
- Task 2: Operating Systems & Scripting for Cybersecurity
- Task 3: Operating Systems & Scripting for Cybersecurity, Network Security & Cryptography Essentials
- Task 4: Operating Systems & Scripting for Cybersecurity
- Task 5: Operating Systems & Scripting for Cybersecurity, Network Security & Cryptography Essentials
- Task 6: Network Security & Cryptography Essentials
- Task 7: Operating Systems & Scripting for Cybersecurity
- Task 8: AI Foundations for Cybersecurity Professionals, Network Security & Cryptography Essentials
- Task 9: Ethical Hacking: Reconnaissance, Scanning & Enumeration, AI-Enhanced Red Team Tactics, and Exploitation to Post-Exploitation
- Task 10: Ethical Hacking: Reconnaissance, Scanning & Enumeration, AI-Enhanced Red Team Tactics, and Exploitation to Post-Exploitation
- Task 11: Ethical Hacking, Reconnaissance, Scanning & Enumeration

Output Screenshots

The security configurations are successfully implemented, monitoring logon attempts and generating real-time reports for both successful and failed logins as shown below:



```
PsLoglist v2.82 - local and remote event log viewer
Copyright (C) 2000-2023 Mark Russinovich
Sysinternals - www.sysinternals.com

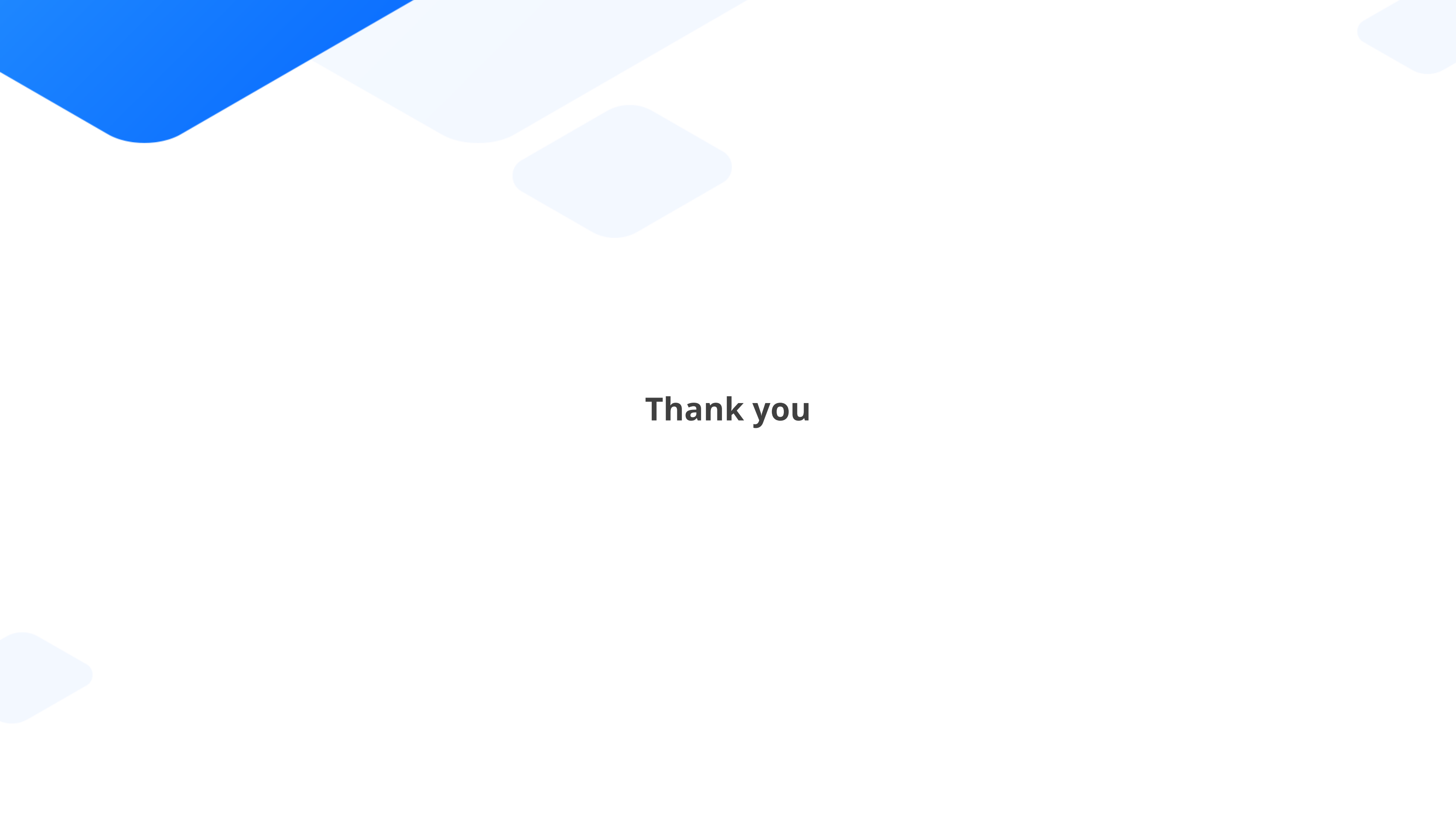
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DOCUMENTATION
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Thank you